

PRS Analytical Pricing Model (MKM-PR-001)

Wolfram Oracle Benchmark of the Fair-Spread Implementation

MKM Research Labs

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1 Executive Summary

This document benchmarks the PRS Analytical Pricing Model (MKM-PR-001): the function that turns a gauge's annual trigger probability into the fair spread of a Physical Risk Swap. It asks two questions, and answers each against references that share none of the model's numerics.

1. **Does the code compute the documented formula?** The model document (`prs_pricing.pdf`, §3.1) specifies the spread exactly. Over 160 cases — probabilities from 10^{-4} to the cap, tenors of 1 to 10 years, two recoveries, a flat rate and the book's yield curve — the model agrees with a 50-digit implementation of that formula to 2.4×10^{-15} (tolerance 1.0×10^{-9}). Wolfram and the 50-digit reference agree to 4.0×10^{-30}, so the reference is confirmed by a second, independent implementation. The floor, the zero rule, the cap and the curve interpolation behave as documented.
2. **Does the model document describe the model accurately?** Three of its statements are numerical, and the benchmark tests them as written (Table 3). Two are not met; the third is a finding.

Findings. The implementation is correct. The documentation is not:

1. **Discretisation is understated (§8.3).** The quarterly scheme prices every spread higher than continuous time by a factor of about $1 + (\lambda + r)\Delta t/2$, at every tenor. That is 75.5 bp at worst for probabilities up to 0.20, where the document says the scheme is accurate, and 675 bp above, where it says "up to 2–3 bp". At UK flood probabilities (around 1–2%) the overstatement is about a bp (Table 4).
2. **The curve is no longer flat (§8.4).** The document says the discount curve is flat; the book prices on a term structure. The effect is below 1 bp for probabilities under 0.1, as the document says, but reaches 9.3 bp.
3. **The basis compares two different formulae (§3.3).** The basis code takes the gauge leg as $p \times 10^4$ and the property leg from Eq. 7. At the same probability the two legs differ, so part of every reported basis is an artefact of the formulae: over 1 bp from $p = 0.01$.

Evidence. 7 checks, outcome: all pass (Table 1).

- Wolfram: 15.0.0 for Mac OS X ARM (64-bit) (May 26, 2026) via live kernel session (wolframclient).
- Code under test: `src/models/hazard/prs_analytical.py` at commit `7bbb63078a59`.
- SHA-256 of that file: `a70f65faf22c369c`...
- Numerics: numpy 1.26.4, scipy 1.15.1, mpmath 1.4.1.
- Generated 2026-10-19 by `python -m tools.wolfram_oracle pr001`.

2 Model Purpose and Scope

2.1 What is benchmarked

`compute_prs_spread` and `interpolate_yield_rate` in `src/models/hazard/prs_analytical.py`: the fair spread every PRS in the book is priced at, and the curve it is discounted on. The module is loaded from its file alone, and called exactly as the book calls it.

2.2 What is not

The basis waterfall's bookkeeping (counts, distances, inverse-distance weights) is not a calculation a mathematical oracle can check, and is out of scope; only the consistency of its two spread legs is examined. The delta engine (MKM-DE-001) and the XVA overlay (MKM-XV-001) consume this function and are benchmarked separately. This is a benchmark of the implementation (validation question Q8), not independent validation (Q9).

3 Mathematical Framework

Following the model document (§3.1): the annual probability p, capped at 0.999, gives the hazard rate $\lambda = -\ln(1-p)$; survival is $S(t) = e^{-\lambda t}$ and discounting $D(t) = e^{-r(t)\,t}$. With quarterly dates $t_i = i\,\Delta t$, $\Delta t = \frac{1}{4}$, $n = 4T$,

$$A = \sum_{i=1}^{n} \Delta t\, S(t_i)\, D(t_i), \qquad P = \sum_{i=1}^{n} (1-R)\big(S(t_{i-1}) - S(t_i)\big)\, D\Big(t_i - \tfrac{\Delta t}{2}\Big), \tag{1}$$

and the fair spread is $s = 10^4\, P/A$ bp, floored at 2 bp. The rate $r(t)$ is either flat or the book's curve, linear between whole-year points and flat beyond them.

Continuous time. With protection paid when the trigger occurs and premium paid continuously, both legs integrate the same $S(t)D(t)$:

$$s_{\mathrm{c}} = 10^4\, \frac{(1-R)\int_0^T \lambda\, S(t)D(t)\,\mathrm{d}t}{\int_0^T S(t)D(t)\,\mathrm{d}t} = 10^4\,(1-R)\,\lambda, \tag{2}$$

whatever the curve and tenor. This is the benchmark the document's §8.3 refers to.

The quarterly scheme in closed form. On a flat curve, $S(t_{i-1}) - S(t_i) = (e^{\lambda\Delta t} - 1)S(t_i)$ and $D(t_i - \Delta t/2) = e^{r\Delta t/2} D(t_i)$, so the sums cancel:

$$s = 10^4\,(1-R)\,\frac{e^{\lambda\Delta t} - 1}{\Delta t}\, e^{r\Delta t/2}, \qquad \frac{s}{s_{\mathrm{c}}} \approx 1 + \frac{(\lambda + r)\,\Delta t}{2}. \tag{3}$$

The premium is paid in arrears with no accrual on a trigger, and the protection half a period earlier; both raise the spread. The fallback of §3.1.7 can never apply: $A \geq \Delta t\, S(T)D(T) > 0$ for any p up to the cap.

4 Benchmark Method

4.1 References

The equations above are evaluated with `mpmath` at 50 significant digits, implemented from the model document rather than the code. Wolfram evaluates the same sums exactly, with the curve as a `Piecewise` function, from each input's exact value at 60-digit precision, returning 30 digits.

4.2 Cases

Ten probabilities (10^{-4} to 0.999), four tenors (1, 3, 5, 10 years), two recoveries (0, 0.4) and two curves (flat 3%, the book's `DEFAULT_YIELD_CURVE`): 160 cases. The book trades $R = 0$ (§3.2); 0.4 checks the recovery term.

5 Validation and Backtesting

Table 1: Benchmark checks

Check	Result	Tolerance	Outcome
Fair spread, all cases (Eqs. 1–8)	2.4×10^{-15}	1.0×10^{-9}	pass
Wolfram agrees with the reference	4.0×10^{-30}	1.0×10^{-9}	pass
Model agrees with Wolfram	2.4×10^{-15}	1.0×10^{-9}	pass
A zero probability prices at zero (s5.3.1)	0 bp, no floor	–	pass
The floor binds below 2 bp (Eq. 8)	p = 0.0001 priced at the floor	–	pass
Probabilities above 0.999 price as 0.999 (s3.1.1)	p = 0.9995 prices as p = 0.999	–	pass
Yield curve: linear between years, flat beyond	max difference 0.0e+00 over 81 times	–	pass

5.1 Fair spread

Table 2: Fair spread on the book curve, $R = 0$: model against reference. The full grid (both curves, both recoveries) is in `spreads.csv`

T	p	Reference spread (bp)	Model rel. err	Model vs W.
1	0.0001	1.00444736302943 (floor)	0	0
1	0.001	10.0501279583779	3.5×10^{-16}	3.3×10^{-16}
1	0.005	50.3767552131246	1.2×10^{-15}	1.2×10^{-15}
1	0.01	101.070945427314	8.8×10^{-16}	8.5×10^{-16}
1	0.02	203.426165376566	1.0×10^{-15}	9.6×10^{-16}
1	0.05	518.499278570660	8.1×10^{-16}	7.4×10^{-16}
1	0.1	1072.28484224664	2.7×10^{-16}	2.4×10^{-16}
1	0.2	2304.91250322506	4.4×10^{-16}	4.5×10^{-16}
1	0.5	7601.46838182698	1.1×10^{-16}	1.3×10^{-16}
1	0.999	185747.399879538	6.4×10^{-17}	4.6×10^{-17}
3	0.0001	1.00546574666302 (floor)	0	0
3	0.001	10.0603116235579	2.0×10^{-16}	1.9×10^{-16}
3	0.005	50.4276694658987	9.1×10^{-16}	9.3×10^{-16}
3	0.01	101.172762500286	6.4×10^{-16}	6.2×10^{-16}
3	0.02	203.629745311317	1.7×10^{-15}	1.8×10^{-15}
3	0.05	519.007644910989	7.1×10^{-16}	6.1×10^{-16}
3	0.1	1073.29834699770	4.7×10^{-16}	5.1×10^{-16}
3	0.2	2306.91521795683	6.0×10^{-16}	5.3×10^{-16}
3	0.5	7605.94277197570	1.6×10^{-16}	1.8×10^{-16}
3	0.999	185747.565606826	6.5×10^{-17}	5.4×10^{-17}
5	0.0001	1.00511329809846 (floor)	0	0
5	0.001	10.0567958313271	2.2×10^{-15}	2.3×10^{-15}
5	0.005	50.4102837185495	6.0×10^{-16}	6.1×10^{-16}

Table 2: Fair spread on the book curve, $R = 0$: model against reference. The full grid (both curves, both recoveries) is in `spreads.csv`

T	p	Reference spread (bp)	Model rel. err	Model vs W.
5	0.01	101.138474129454	1.6×10^{-15}	1.6×10^{-15}
5	0.02	203.563101221979	9.1×10^{-16}	8.6×10^{-16}
5	0.05	518.855511123588	8.3×10^{-16}	8.6×10^{-16}
5	0.1	1073.04190294206	7.9×10^{-17}	6.9×10^{-17}
5	0.2	2306.58521785897	2.7×10^{-16}	2.0×10^{-16}
5	0.5	7606.04114076090	4.6×10^{-17}	2.2×10^{-17}
5	0.999	185747.565607130	1.7×10^{-16}	1.8×10^{-16}
10	0.0001	1.00509617460275 (floor)	0	0
10	0.001	10.0566227760028	2.2×10^{-15}	2.1×10^{-15}
10	0.005	50.4093789121315	4.7×10^{-16}	4.7×10^{-16}
10	0.01	101.136569926700	9.3×10^{-16}	9.6×10^{-16}
10	0.02	203.558942620238	8.2×10^{-16}	8.4×10^{-16}
10	0.05	518.843044552095	4.5×10^{-16}	4.9×10^{-16}
10	0.1	1073.01359166405	2.1×10^{-16}	2.1×10^{-16}
10	0.2	2306.53611328221	2.7×10^{-16}	2.7×10^{-16}
10	0.5	7606.04561980318	1.6×10^{-16}	1.1×10^{-16}
10	0.999	185747.565607130	1.7×10^{-16}	1.8×10^{-16}

5.2 The model document's claims

Table 3: The model document's claims, tested

Claim	Document says	Benchmark finds	Outcome
s8.3 Quarterly vs continuous time	up to 3 bp, only above $p = 0.20$	75.5 bp ($p \leq 0.20$); 674.6 bp above	**not met**
s8.4 Flat vs book curve, 5y	< 1 bp	9.3 bp; over 1 bp from $p = 0.1$	**not met**
s3.3 Basis: gauge leg $p \times 10^4$ vs Eq. 7	not stated	up to 2606 bp; over 1 bp from $p = 0.01$	finding

Table 4: Worked figures at 5 years, $R = 0$, in bp

p	Quarterly	Continuous	Δ	Rel.	Flat−book	$p \times 10^4$	Basis artefact
0.0001	1.00	1.00	+0.00	+0.38%	-0.00	1.0	-0.01
0.001	10.04	10.01	+0.04	+0.39%	-0.01	10.0	-0.06
0.005	50.35	50.13	+0.22	+0.44%	-0.07	50.0	-0.41
0.01	101.01	100.50	+0.50	+0.50%	-0.13	100.0	-1.14
0.02	203.30	202.03	+1.27	+0.63%	-0.26	200.0	-3.56
0.05	518.18	512.93	+5.24	+1.02%	-0.68	500.0	-18.86
0.1	1071.61	1053.61	+18.01	+1.71%	-1.43	1000.0	-73.04
0.2	2303.47	2231.44	+72.04	+3.23%	-3.11	2000.0	-306.59
0.5	7596.72	6931.47	+665.25	+9.60%	-9.32	5000.0	-2606.04

The relative column of Table 4 follows $(\lambda + r)\Delta t/2$; the book curve's rates sit above 3%, which is why it prices slightly higher than the flat curve.

6 Model Limitations and Known Weaknesses

1. **Benchmark, not validation.** The code is shown to compute the documented formula; whether a constant hazard rate and quarterly pricing suit the product is a separate question.

2. **Documentation.** §8.3 and §8.4 of the model document should be corrected to the figures above, and the code's docstring, which says the protection leg is discounted at $D(t_i)$, to the midpoint the code and the document both use.

3. **Basis legs.** Whether the gauge leg should be priced with Eq. 7, so that the basis measures the property against the gauge rather than one formula against another, is for the model owner.

4. **High probabilities.** Near the cap the spread exceeds 10^5 bp; the formula is computed correctly there, but the number has no economic meaning.

7 Document History

Date	Version	Description	Author
19-Oct-2026	0.1	First issue	David K Kelly